A Geometric Feature–based Conditional Diffusion Model for Predicting Stress and Strain Distributions in Curved OLED Screens

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Abstract

Finite element simulation (FES) is generally employed to predict the stress and strain distributions of curved OLED screens; however, this method is time-consuming and often requires several hours to complete. This paper proposes a geometric feature–based conditional diffusion model (CDM) that enables rapid high-precision prediction of stress and strain for various cover glass shapes of curved OLED screens. Experimental results demonstrate that the proposed method makes second-scale predictions of such stress and strain distributions, achieving >93% of the prediction accuracy of FES.

Author Keywords

Finite Element Simulation; Stress; Strain; Curved OLED Screens; Geometric Feature; Conditional Diffusion Model

1. Introduction

Curved organic light-emitting diode(OLED) screens consist of stacked structures, including rigid cover glass, a flexible touch panel, OLED device films, and a backplane, where each film layer is bonded using optical adhesives [1-2]. The ductility values of the rigid cover glass, various film layers, and the optical adhesive materials differ significantly. Consequently, residual stresses are generated between layers owing to inconsistent deformation during the lamination process. When the accumulated residual stress exceeds the interfacial bonding strength of the materials, defects such as wrinkling of the flexible display panel and damage to the OLED device films may occur [3]. The finite element simulation (FES) technique is commonly employed to construct three-dimensional (3D) models for stress and strain prediction. However, the full 3D simulation of the lamination process for a curved OLED screen typically requires several hours, significantly hindering design optimization. Therefore, artificial intelligence (AI) surrogate models are needed to accelerate mechanical simulation and enable efficient design−space exploration and optimization. Although traditional machine learning methods have been used for stress and strain prediction, they can typically predict only a limited number of sparse simulation outputs, and lack the capability to predict two-dimensional (2D) stress and strain fields for complex structures exhibiting nonlinear mechanical properties [4-5].

Although deep learning models, such as U-Net [6], are capable of predicting dense 2D simulation output data, conventional deep learning approaches generally produce relatively blurry predictions. Their prediction accuracy tends to be low when the distribution of simulation output data exhibits significant fluctuations. The development of generative AI technologies represents a new possibility for resolving difficulties in stress and strain prediction in complex structures. In addition, existing AI-based simulation prediction models typically assume fixed geometric configurations and therefore; lack adaptability to structures with varying geometries [7, 8]. Conditional diffusion model (CDM) has recently emerged as a research hotspot in the field of nonlinear prediction, owing to their unique ability to stably generate complex distributions and fit high-dimensional data, even when training data are limited [9-11]. This paper proposes a stress–strain prediction method for curved OLED screens using geometric features (GFs) and a CDM. First, a multidimensional dataset covering different curved-surface geometric structures and material property parameters is constructed, and labeled samples are generated via FES. Second, the geometric structural features and material property parameters of the curved OLED screens are extracted as conditional information, and a CDM is developed with a parameter attention module incorporated into the diffusion process to enhance its ability to capture features of key design parameters. This approach provides technical support for rapid

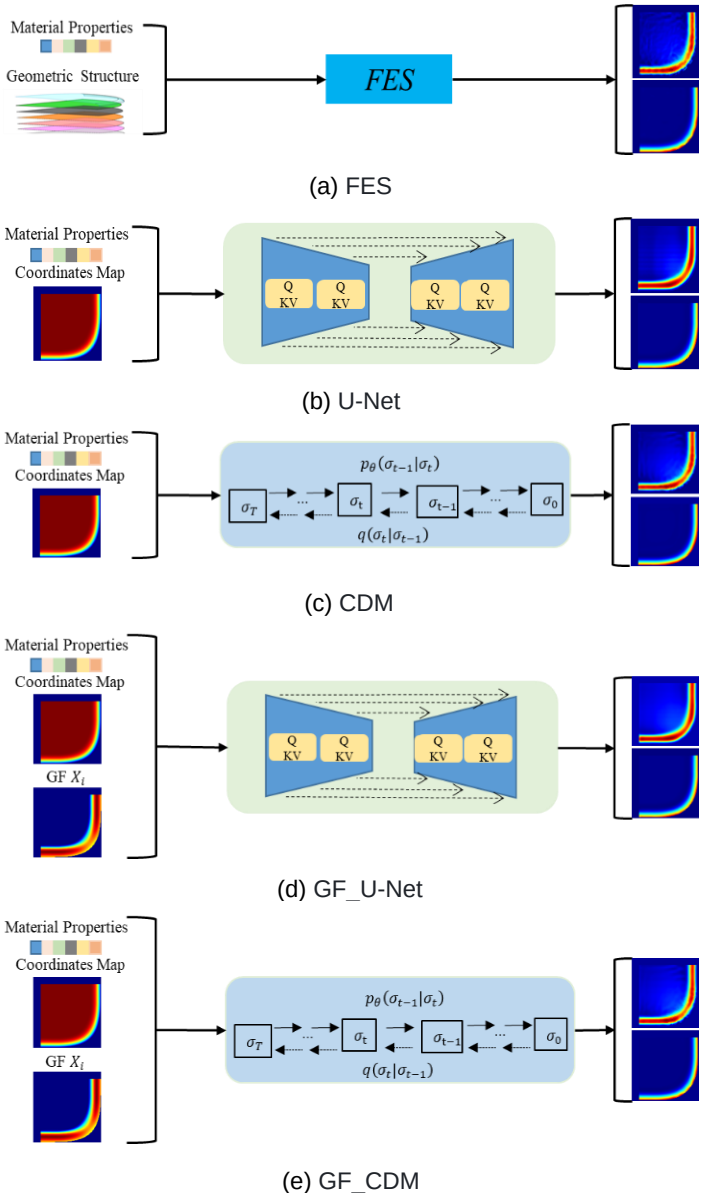


Figure 1. Architecture of the AI surrogate model used for FES.

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product design and process optimization while offering a methodological reference for mechanical property prediction in other complex structures.

2. Proposed Methodology

Given the long computational time required for the FES of curved OLED screens, an AI model was designed to replace FES. Several types of rigid curved-cover-glass configurations were selected. For each cover glass configuration, a hypercube space uniform sampling method was used to generate different parameter combinations that varied the material properties and film thickness of the flexible display panel. Python scripts were used to generate FES models in batch, and FES was performed to produce training samples. An AI surrogate model was designed, which consisted primarily of two components: feature-parameter extraction and the AI prediction model. For feature-parameter extraction, the material property parameters of each layer of the flexible display panel, element coordinates, and GFs of the curved cover glass were extracted. Additionally, a top–down projection was employed to convert the 3D mesh structure into a 2D image, allowing the 3D stress–strain prediction problem in FES be reformulated as a 2D stress–strain field prediction task. After converting the 3D mesh structure to a 2D planar representation, image-based deep learning models end-to-end and image-generation models could be applied. Herein, an end-to-end U-Net model was used as the baseline, and an image-generation CDM was developed in parallel. The system design of the AI surrogate model for FES is illustrated in Figure 1. Figure 1(a) shows the finite element simulation schematic; Figures 1(b)–(e) present four AI surrogate models corresponding to different combinations of inputs and AI models.

Feature Extraction: Owing to confidentiality constraints, the GFs and material property features extracted in this study are collectively denoted as X_i. Using the prediction of the maximum principal stress distribution in the conductive layer and the z-direction deformation distribution in the light-emitting layer of the OLED device thin film as examples, this paper elaborates on the relevant methodology. Notably, the prediction methods for stress and strain distributions in other layers follow the same procedure and are thus will not described individually. The GFs (X_1, X_2, and X_3) of each element in the curved cover glass were calculated. The coordinates (x, y, z) of the elements in each sub-layer were extracted, where z denotes the vertical distance of the current z-coordinate of the sub-layer (at the same (x, y) position) relative to that of the curved cover glass, generating the top-view coordinate map of the current layer. The material properties and thickness parameters of the flexible display panel sub-layers as well as the stress and strain values corresponding to each element were also extracted.

2D Feature Projection: From the top-view perspective, the extracted feature parameters were mapped in a one-to-one manner onto an image with a resolution of 128 × 128 pixels according to the coordinate distribution of the elements in each layer. For pixel locations within the element regions that did not contain corresponding elements, the feature parameter values mapped to these pixels were calculated using linear interpolation from adjacent valid pixel points.

U-Net Model: The U-Net architecture utilized in this study is a modified version of the classic encoder–decoder structure; designed to enhance its ability to model material properties and GFs. This architecture takes 128 × 128 feature maps of material properties and GFs as input; and predicts stress–strain distribution maps through the collaborative interaction between the encoder and decoder paths. The encoder path consists of an initial convolutional layer followed by three downsampling layers. The initial convolutional layer adopts a residual double-convolution module composed of 3 × 3 convolutions, batch normalization, ReLU activation functions, and a dropout layer. Residual connections are incorporated to alleviate the gradient-vanishing problem in deep networks, enabling the effective extraction of basic features while suppressing overfitting. Each downsampling layer further refines the features via a residual double-convolution module (which is also integrated with a dropout layer), followed by max pooling to reduce the feature map size by half. Additionally, each downsampling layer can be equipped with a local self-attention module. This module divides the feature map into dynamic windows; computes attention only within each window; and generates the query (Q), key (K), and value (V) representations through 1 × 1 convolutions. It then performs matrix operations to efficiently capture local spatial correlations. The decoder path comprises three upsampling layers. It restores the feature map size via bilinear interpolation or transposed convolution and concatenates the upsampled features with the corresponding feature maps from the encoder path through skip connections. The concatenated features are processed using the residual double-convolution module with a dropout layer. Additionally, the decoder path can be configured with local self-attention modules to enhance the modeling of feature correlations. Finally, the output layer generates 128 × 128 stress–strain prediction maps. During the training phase, two additional deep supervision auxiliary outputs are used to accelerate convergence. The core advantageous features of this architecture include the following: (i) the dropout layer, which effectively improves model generalizability; (ii) the integration of local self-attention, which accurately captures local correlations between material properties and GFs. This design significantly enhances the capability of the model to represent complex material mechanical properties while maintaining computational efficiency.

CDM: The CDM is a deep learning architecture based on probabilistic generation. By incorporating prior information such as physical constraints and boundary conditions, it transforms the traditional unconstrained diffusion process into a conditional generation task. In the stress and strain prediction of curved laminated screens, this model can integrate the geometric shape of the curved cover glass and the material properties of the flexible display panel to achieve highly accurate and robust predictions of the stress–strain distribution.

The prediction of stress and strain in OLED curved laminated screens based on a CDM involves two sequential processes: the forward noise-adding process and the reverse denoising process [12]. Given an initial real stress–strain field σ_0, the forward diffusion process gradually adds Gaussian noise through a Markov chain, generating a sequence of variable $\{\sigma_t\}_{t=1}^{T}$ with progressively increasing noise levels.

$$\mathrm{q}(\sigma_t|\sigma_{t-1}) = N\left(\sigma_t; \sqrt{1-\beta_t}\sigma_{t-1}, \beta_t I\right) \quad (1)$$

where β_t denotes the noise scheduling parameter; and I represents the identity matrix. Through the reparameterization trick, the state at any time step t can be directly calculated from σ_0 as follows:

$$\sigma_t = \sqrt{\bar{\alpha}_t}\sigma_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \epsilon \sim N(0, I) \quad (2)$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. During the reverse denoising process, the neural network p_θ learns the conditional distribution $\mathrm{q}(\sigma_{t-1}|\sigma_t, \boldsymbol{c})$ to gradually recover the stress–strain field as follows:

$$p_\theta(\sigma_{t-1}|\sigma_t, \boldsymbol{c}) = N(\sigma_{t-1}; \mu_\theta(\sigma_t, t, \boldsymbol{c}), \sigma_t^2 \boldsymbol{I}) \quad (3)$$

The mean function μ_θ is parameterized using the predicted noise:

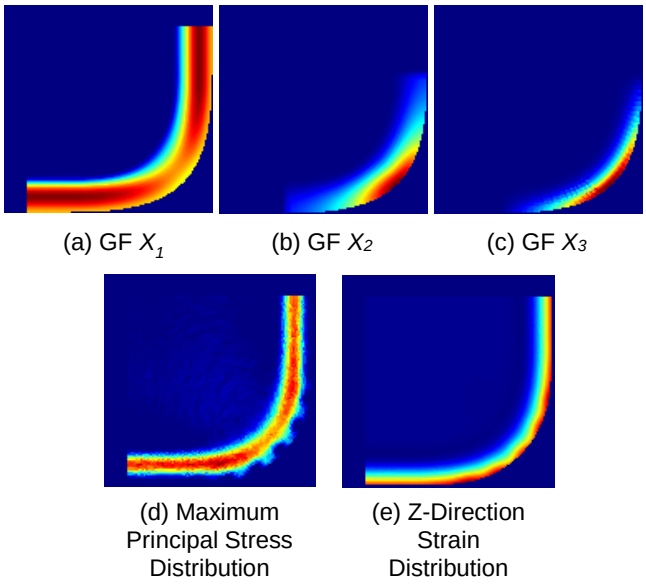
$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}\left(\sigma_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\sigma_t, t, \boldsymbol{c})\right) \quad (4)$$

where ϵ_θ denotes the noise predicted by the neural network, and $\boldsymbol{c}$ represents conditional parameters.

The reverse denoising network adopts a U-Net architecture. To achieve the deep fusion of multi-source conditions with the diffusion process, time-step embedding converts discrete time steps (t) into high-dimensional continuous features, capturing differences in noise distributions across different noising stages. The network learns the noise distribution at a time step t based on conditional information. The training objective is to minimize the difference between $p_\theta(x_{t-1}|x_t)$ and $q(x_{t-1}|x_0)$, with a mean squared error loss function used to supervise model training.

3. Experimental Results

Datasets: The test set was regenerated using an FES software. In particular, OLED screens with curved cover glass of varying shapes were generated to verify the model generalizability to other structures of the curved cover glass. Meanwhile, some simulation validation sets with inconsistent z-direction deformation at the screen edges were generated to verify the ability of the model to predict wrinkles.



(a) GF X_1 (b) GF X_2 (c) GF X_3

(d) Maximum Principal Stress Distribution (e) Z-Direction Strain Distribution

Figure 2. Geometric Features and Stress–Strain Projection.

Curved OLED screens consist of two main components: the rigid curved cover glass and the flexible display panel. For mesh division, the curved cover glass was modeled using quadrilateral shell elements, and the sub-layers of the flexible display panel are meshed with structured hexahedral element. An FES software was used to simulate the stress in the conductive layer and the z-direction deformation of the light-emitting layer within the OLED device film. The total runtime consumption of this simulation model is between 9 and 10 h.

2D Feature Projection Results: Figures 2(a)-(c) show the 2D projection diagrams of the GFs (X_1, X_2, and X_3) of the curved cover glass after feature extraction and projection. The maximum principal stress distribution of the conductive layer and the z-direction strain distribution of the light-emitting layer are shown in Figures 2(d) and (e).

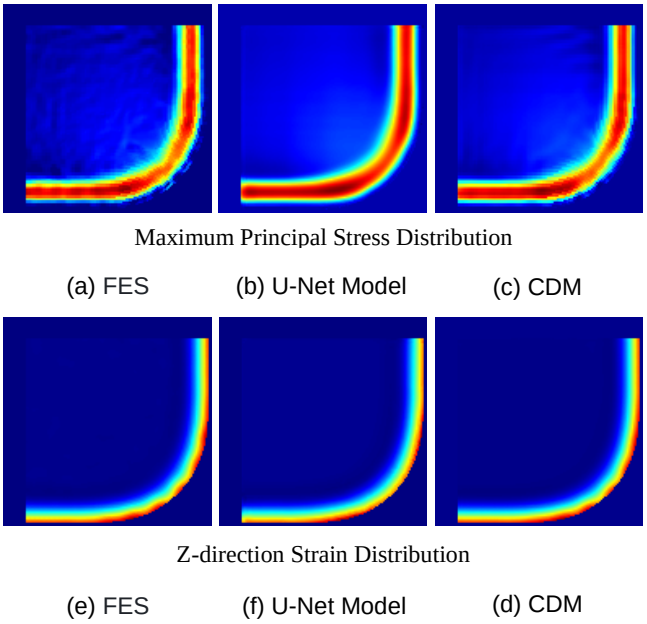
Evaluation Metrics: The mean relative error (MRE) quantified the overall error between the predictions of the deep learning model and the stress–strain distributions obtained from the FES. For curved laminated screens, particular attention is given to the maximum stress value and the peak-to-peak z-direction deformation at the screen edge (which characterizes wrinkles). Therefore, the peak stress value was extracted to calculate the peak relative error (PRE) for evaluating the prediction accuracy of the stress field. Further, the peak-to-peak z-direction deformation at the screen edge was extracted to calculate the peak-to-peak relative error (PPRE) for evaluating the prediction accuracy of the strain field.

$$MRE = \left|\frac{\bar{S}_{DL} - \bar{S}_{FES}}{\bar{S}_{FES}}\right| \times 100\% \quad (5)$$

$$PRE = \frac{|\max(|S_{DL}|) - \max(|S_{FES}|)|}{\max(|S_{FES}|)} \times 100\% \quad (6)$$

$$PPRE = \sum_{i=1}^{N} \frac{|PP_{FES} - PP_{DL}|}{|PP_{FES}|} \times \frac{1}{N} \times 100\% \quad (7)$$

In Equations 5 and 6, S_{FES} and S_{DL} represent the stress and strain distributions obtained from FES and the deep learning model, respectively. In Equation 7, PP denotes the local peak-to-peak value of the stress–strain field, which is calculated by subtracting the local trough from the local peak.



Maximum Principal Stress Distribution

(a) FES (b) U-Net Model (c) CDM

Z-direction Strain Distribution

(e) FES (f) U-Net Model (d) CDM

Figure 3. Stress and Strain Distributions Determined Using FES Method and Those Predicted by the GF_U-Net and GF_CDM Models.

Prediction Results: This study also used the U-Net model for stress and strain prediction, with the U-Net architecture was being consistent with that of the CDM. Figure 3 shows the stress and strain distributions predicted by FES, the GFs-guided U-Net model, and the CDM. The stress and strain distributions predicted by the two AI surrogate models generally aligned closely with FES results, with the CDM showing greater consistency with the FES results in terms of details. Table 1 summarizes the evaluation metrics for the prediction results of the CDM and U-Net model with

and without GFs guidance. The CDM trained on a small-scale dataset using the extracted GFs of the curved cover glass outperformed the U-Net model in both evaluation metrics on the curved-screen test set with different curved structures, achieving values <7% for both metrics. Meanwhile, models trained without integrating curved cover-glass GFs exhibited significantly low evaluation metric values on the same test set. Specifically, the MRE for z-direction strain distribution predicted by the CDM reduced by ≈22%. This indicated that when GFs of the curved cover glass were not considered, the CDM struggled to reconstruct strain distributions from random noise. Notably, when GFs of the curved cover glass were used, The CDM exhibited lower PPRE in strain prediction. This was because the strain distributions contained peaks and valleys, and the conditional diffusion process gradually reconstructed the local fine details of strain distributions. Conversely, the U-Net model tended to homogenize these peaks and valleys to reduce the overall loss function. Table 2 presents the time consumption statistics for stress–strain prediction using the three models: FES, U-Net, and the CDM. The AI surrogate model proposed in this study significantly enhanced the efficiency of mechanical simulations.

Table 1. Comparison of Stress and Strain Prediction Results of U-Net and the CDM (with and without GFs).

Model	Maximum Principal Stress		Z-direction Strain	
	MRE	PRE	MRE	PPRE(edge)
U-Net	9.5%	8.65%	9.23%	37.54%
CDM	8.62%	7.76%	23.36%	7.67%
GF_U-Net	6.98%	7.07%	1.94%	11.57%
GF_CDM	**6.78%**	**6.19%**	**0.45%**	**4.63%**

Table 2. Time Consumption for Stress and Strain Predictions Using FES, U-Net, and the CDM.

	FES	U-Net	CDM	GF_U-Net	GF_CDM
Time(s)	33664	1.8	8.5	1.9	8.8

4. Discussion

This article proposes a GF–based CDM for predicting stress and strain distributions in curved OLED screens. This method enables stress–strain distribution predictions within seconds, with a prediction accuracy >93% of FES. Specifically, GFs of the rigid curved cover glass in curved OLED screens are extracted and used them as one of the inputs to the CDM. This ensures that the AI model maintains high prediction accuracy even for test sets with variable geometric structures of OLED curved surfaces. This significantly expands the application scenarios of the proposed model for stress–strain prediction in curved screens. Furthermore, compared with the conventional end-to-end U-Net model, the CDM developed in this study exhibits superior prediction accuracy and detail fidelity pertaining to stress–strain characterization. Experimental results on the computational time taken by the AI surrogate model confirm that this method substantially improves the efficiency of stress–strain simulation and prediction. Consequently, this method holds great potential as a fundamental tool for the optimizing geometric structures and material selection in curved OLED screens. By integrating global and local parameter optimization approaches, the optimal design of curved OLED screens can be realized, providing a valuable reference for developing surrogate models for FES-based structural mechanics in other complex systems.

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